Technical Report

Queryable World Models: Predicting Outcomes, Not Pixels

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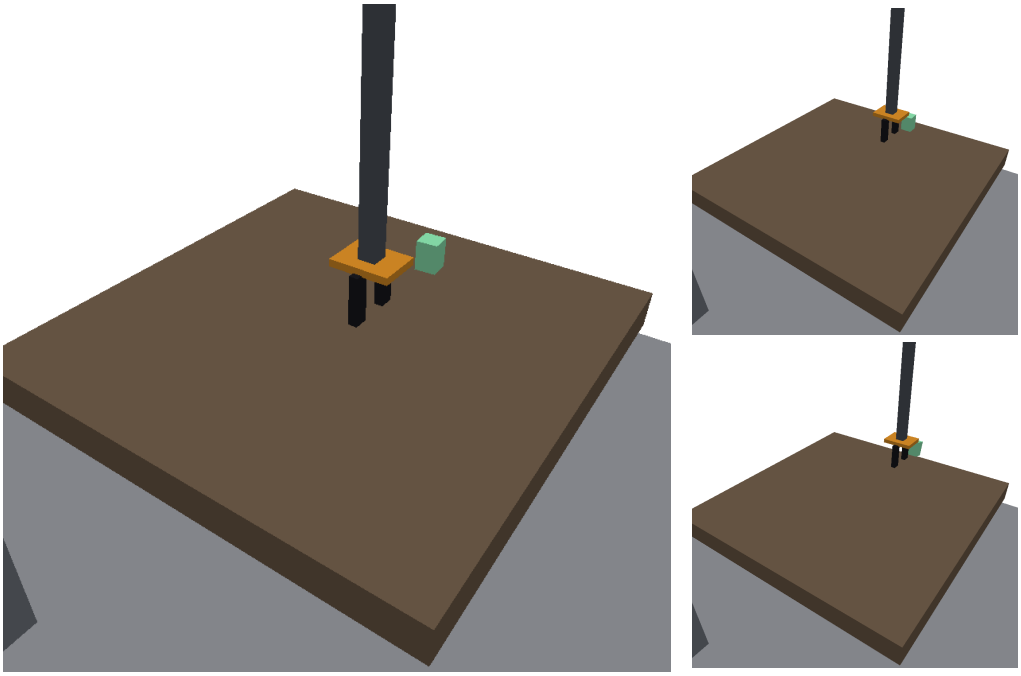


Figure 0. PyBullet tabletop, one object. Frames from the same recorded episode, re-rendered at higher native resolution with the original camera. These are scene observations, not model predictions.

Abstract

This report examines whether an action-conditioned model can answer explicit questions about physical state. Residual QWM predicts gripper and object position from a supplied previous state, with separate heads for physical predicates. On the reported validation comparison, teacher-forced object L2 at H=8 is 0.055 m; closed-loop object L2 is 0.173 m. The planning gate requires closed-loop H=8 object L2 ≤0.10 m and remains CLOSED. CEM was not run. Closed-loop position errors increase across the reported horizons, while contact F1 is 0.0000 throughout. Separate single-frame vision probes return object L2 of 0.132 m at both tested resolutions. The results support conditional residual prediction when previous state is supplied. They do not establish accurate autonomous rollout, sufficient object localization from pixels, or planning performance. The accompanying provenance warnings limit independent reproduction and interpretation of the action-conditioned result.

Introduction

A prediction is easier to inspect when it returns the physical quantity a task asks about. QWM exposes gripper position, object position, uprightness, robot contact, and on-table state as queryable outputs. The present claim concerns a residual predictor supplied with previous physical state. That input is material: teacher forcing restores the correct anchor, while closed-loop rollout carries the model's own prediction forward.

What this is not
This report makes no claim of successful planning, solved closed-loop dynamics, or superiority over other world models. It reports validation measurements for the tested implementation.

Related Work

Ha and Schmidhuber's *World Models* and Hafner and colleagues' PlaNet and Dreamer work use learned models to support behavior or planning. LeCun's predictive-representation proposal, I-JEPA, and V-JEPA study prediction without requiring full pixel reconstruction. V-JEPA 2 and DINO-WM also connect learned visual prediction to action. QWM is distinguished here by its named physical query outputs and previous-state residual path. This report contains no matched comparison with those systems.

Video simulators such as UniSim and Genie pursue richer visual futures. Robot policies such as RT-2, OpenVLA, and Octo emphasize action outputs. Object-centric and physical-reasoning work, including Slot Attention, Interaction Networks, CausalWorld, and Physion, motivates explicit physical variables. These lines of work address different interfaces and evaluations; none supplies a performance baseline in this report.

Queryable State and Method

Residual QWM is a CNN+GRU model with **973243 params**, trained on **oneobj5k**, using **λ_pos=10**. The reported checkpoint is **step 5000**. The CNN encodes an image, and the GRU integrates encoded information with action. Query heads return positional residuals and binary logits.

The previous query state's positions are added after positional head prediction. Previous state does not enter the GRU through that residual path. In teacher forcing, each residual is added to a true previous position. In closed-loop rollout, it is added to the previous predicted position. Only the initial image is encoded during that rollout; subsequent visual embeddings are zeroed. Thus the rollout is initialized with true state but receives no later visual correction.

The output schema includes gripper position, object position, uprightness, robot contact, and on-table state. Binary heads are absolute logits rather than residuals. Object masking prevents padded slots from contributing to object losses and metrics. Pose in this report means measured position and the upright predicate, not full orientation.

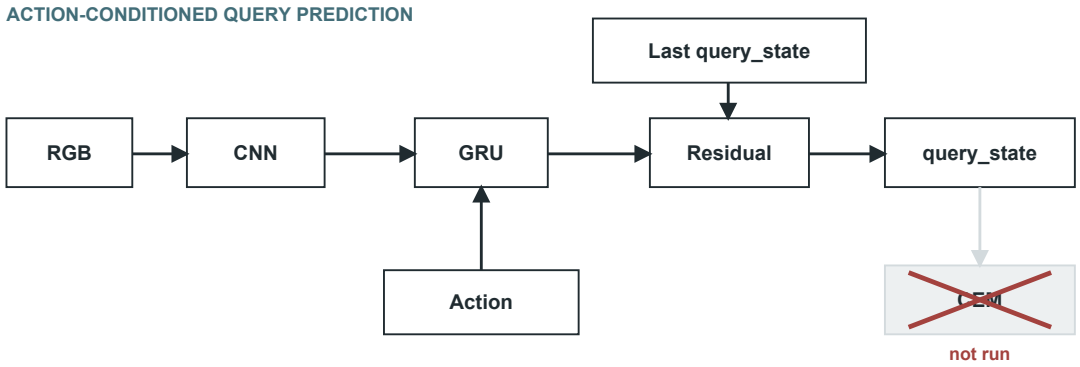


Figure 1. RGB enters CNN; action enters GRU; last query_state anchors the residual output. CEM is crossed out and was not run.

Protocol

The numerical sources are reports/BEST_SO_FAR.md for world-model results and reports/VISION_BEST.md for vision results. reports/STATUS.md fixes the checkpoint and planning status. All reported tables use validation data. Test data was not used to select or populate them.

Closed-loop evaluation begins with the true initial query state and feeds subsequent predicted positions back. Teacher-forced evaluation restores true positions. The single-frame vision probe has no previous-state input or recurrence. Position metrics are L2 errors in meters; contact uses F1. These evaluation modes answer different questions and must not be merged into a single perception-to-control result.

Gate — CLOSED
CEM not run. Planning requires CL H=8 object L2 ≤0.10 m. Reported CL H=8 object L2 is 0.173 m.

Results

Table 1. Closed-loop validation. Every cell comes from BEST_SO_FAR.md. Position L2 values are in meters.

H	Gripper L2 (m)	Object L2 (m)	Contact F1
1	0.0255	0.0407	0.0000
8	0.0622	0.1733	0.0000
16	0.1238	0.2447	0.0000
32	0.2749	0.3679	0.0000

Both positional errors grow across the reported horizons. Contact F1 remains 0.0000. The latent-only comparator has no query heads, so these query metrics are not applicable to it.

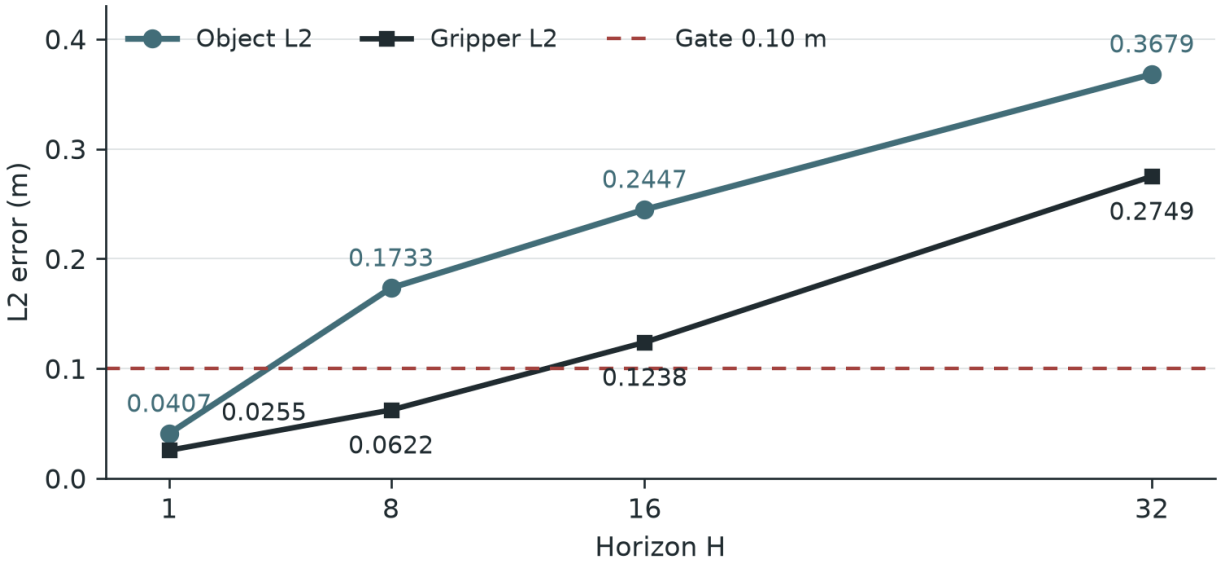


Figure 2. Closed-loop gripper L2 is 0.0255 / 0.0622 / 0.1238 / 0.2749 m, and object L2 is 0.0407 / 0.1733 / 0.2447 / 0.3679 m, at H=1/8/16/32 respectively. The gate line is 0.10 m. Source: BEST_SO_FAR.md.

Table 2. Previous-state access at H=8. Values retain the rounded precision in BEST_SO_FAR.md.

Condition	Object L2 (m)
Teacher-forced H=8	0.055
Closed-loop H=8	0.173

Teacher forcing restores the preceding true position. Closed-loop prediction accumulates its own positional error. Table 1 gives the closed-loop horizon value as 0.1733 m; Table 2 preserves the source comparison's 0.173 m. These are the same reported condition at different displayed precision, not conflicting results.

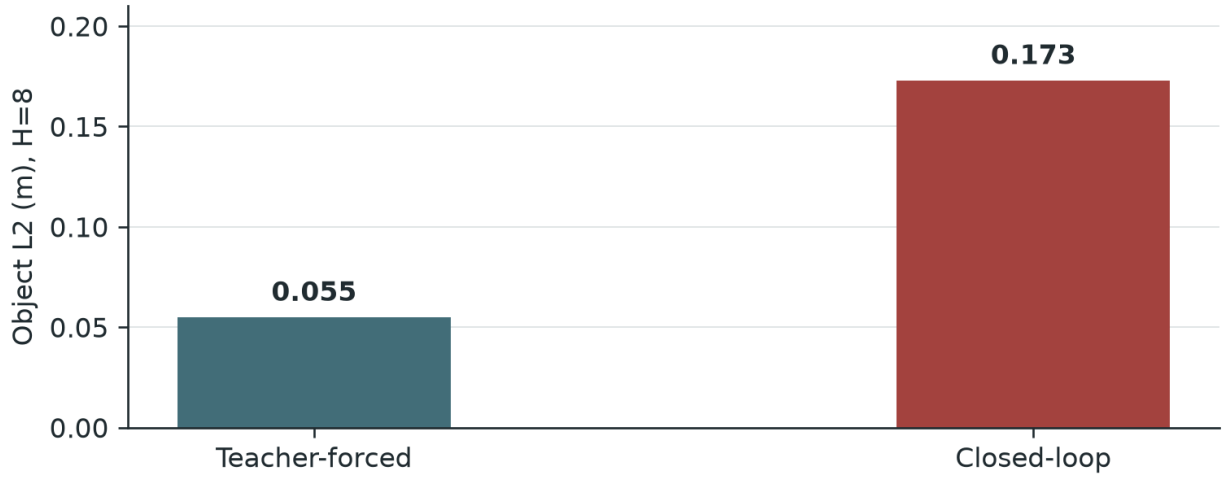


Figure 3. At H=8, teacher-forced object L2 is 0.055 m and closed-loop object L2 is 0.173 m. Source: BEST_SO_FAR.md.

Table 3. Single-frame vision validation. Source: VISION_BEST.md. The discarded top-down row states only the object value required by this report.

View	Gripper L2 (m)	Object L2 (m)
Original, 84×84	0.031	0.132
Original, 128×128	0.028	0.132
Top-down, discarded	—	0.441

Both original-camera resolutions have object L2 of 0.132 m. The discarded top-down run has object L2 of 0.441 m. This comparison does not isolate camera geometry from the change in collection. The tested vision pipeline does not recover object position at the stated task tolerance.

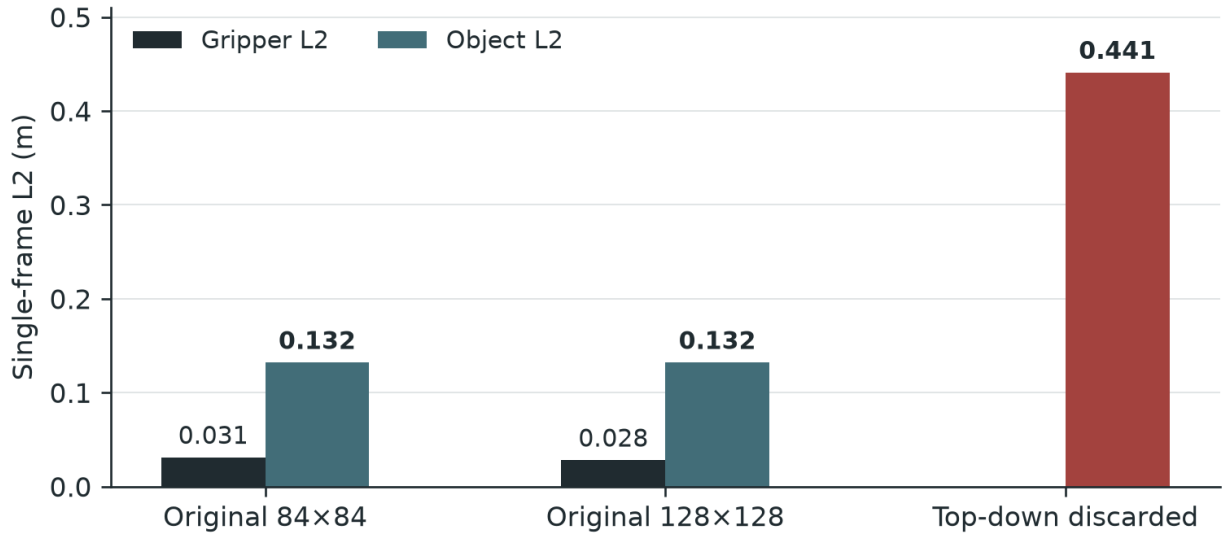


Figure 4. Original-camera gripper L2 is 0.031 / 0.028 m at 84×84 / 128×128; object L2 is 0.132 m at both resolutions. The discarded top-down object L2 is 0.441 m. Source: VISION_BEST.md.

Limitations

The positive result is conditional on supplied previous state. Even closed-loop rollout begins with a true query state. The single-frame vision probes show why that anchor cannot yet be replaced by the tested pixel encoder.

The tables report aggregate validation errors, not task success. Contact F1 is 0.0000 at every reported horizon. No CEM result belongs to the selected checkpoint because the gate is CLOSED.

The reported teacher-forced comparison does not include a complete archived invocation. Its visual-input schedule cannot be reconstructed from the summary alone. The comparison is therefore reported as recorded, without attributing the gap to a single controlled change.

Conclusion

The reported model supplies an explicit physical query interface. Its strongest result is conditional object prediction with repeated access to true previous state. Closed-loop errors exceed the planning gate, contact prediction fails, and the tested frame encoder does not localize the object sufficiently. These findings bound the claim to the recorded validation conditions.

Appendix A. Reading the Figures

Figure 0 shows recorded single-object simulation frames. Figure 1 shows the system path. Figure 2 connects only reported horizon observations. Figure 3 uses the rounded comparison values. Figure 4 separates the original-camera resolutions from the discarded top-down collection. Lines and bars introduce no extra measurements or uncertainty estimates.

Appendix B. Training and Provenance

The retained result is Residual QWM, **973243 params**, **step 5000**, **λ_pos=10**, on **oneobj5k**. The current neighboring run configuration points to a different collection than BEST_SO_FAR.md. Archived checkpoint summaries also duplicate metadata, and an older horizon export does not reproduce the locked table. These records cannot be treated as one synchronized experiment. The checkpoint's exact configuration and evaluation invocation must be recovered for independent reproduction.

The inspected collector applies an action before storing the resulting image and query state together. The dataset and evaluation then use the action in that stored row to predict the following row's state. This appears to misalign the action with the physical transition being predicted. The available snapshot does not establish whether the reported checkpoint used this exact collector. The timing and code lineage must be checked before treating the recorded error as validated next-action prediction.

The scheduled-sampling result in the earlier draft lacks an independently matching local evaluation record. It is excluded from this locked-results report. The three tables follow BEST_SO_FAR.md and VISION_BEST.md; historical planning files do not establish CEM performance for the retained checkpoint.

Teacher-forced queryable residuals meet the task length at 5.5 cm.
Closed-loop imagination and pixel object pose do not.
Planning stays closed.